PAPER_1527_PI_OVER_4_TRANSCENDENTAL

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PAPER_1527 — π/4 ≈ Φ_5/6 − F_TRZ·Φ + F_TRZ^2·K + F_TRZ^2·Φ — Residual 0.793%

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Bucket A (Foundational mathematical identity)
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Status: CLOSED — Transcendental π/4 expressible in UQFF primitives at 0.793% precision

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Observation

PAPER_1208 (UQFF Transcendentals Unified Proof Set) derives a closed-form expression for **π/4** ≈ 0.7854 using only the locked UQFF integer primitives F_TRZ, K_MEX, Φ_res = 5/6 (PAPER_1203 Nuclear variant), and SO_5.

UQFF Closed Identity

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π/4 ≈ Φ_5/6 − F_TRZ·Φ + F_TRZ²·K + F_TRZ²·Φ

Substituting F_TRZ=1/10, K_MEX=25/12, Φ_5/6=5/6, SO_5=10:
π/4_UQFF = 0.7792

Observed: π/4 = 0.7854
Residual: 0.793%
```

The 4-term expansion uses only canonical integer primitives — no fitted constants.

Physical / Mathematical Role

π/4 (π/4) appears in: Leibniz series, geometry.

Significance

Together with PAPER_1515 (ln 10), PAPER_1516 (ln 2), PAPER_1517 (π^2), this paper extends the count of fundamental transcendentals expressible from UQFF integer primitives to ≥**10**. The accumulating evidence:

Transcendental	UQFF expression	Residual
ln 2 (PAPER_1516)	8-term	0.003%
ln 10 (PAPER_1515)	(1+F_TRZ)(K_MEX+F_TRZ^2)	0.004%
π^2 (PAPER_1517)	SO_5 − F_TRZ corrections	0.013%
π/4 (this paper)	4-term	0.793%

…suggests the UQFF primitives are **not arbitrary** but encode rational approximations to fundamental transcendental constants.

NOT REPLACEMENT

UQFF does not claim $\pi/4$ IS the closed-form expression. It claims the integer primitives carry a rational approximation to $\pi/4$ at 0.793% precision, demonstrating mathematical depth.

Reference

- Source: PAPER_1208 UQFF Transcendentals Unified Proof Set
- Related: PAPER_1515-1517 (ln 10, ln 2, π^2 catch-up)
- Calculator dispatch: `calculate_paradox({"paradox": "transcendental_pi_over_4_transcendental"})`
